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D21 eStation Developer Manual

This document applies only to ETAG ESL Gen 3.0 and DSL Gen 1.0 system integration.
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1 Introduction

1.1 Background

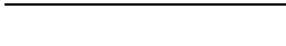
The eStation IoT device, type code AP05, is designed for system integrators to rapidly develop their business projects using the MQTT v5 protocol.

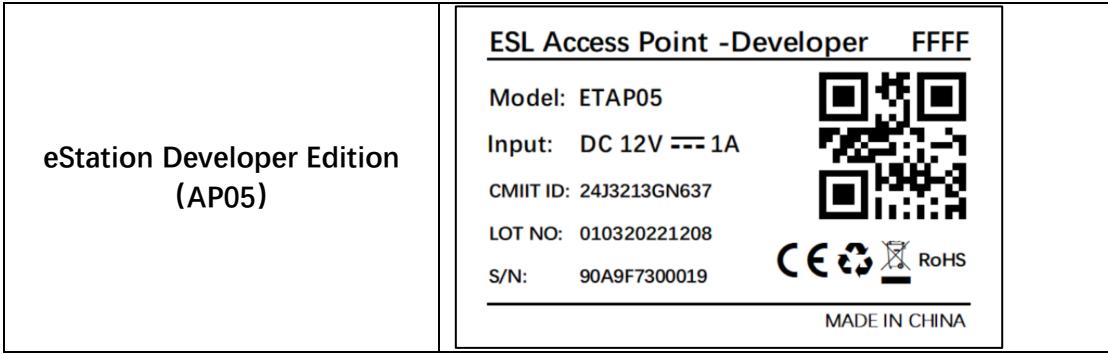
Difference between AP04 and AP05: Before using eStation, please carefully understand the differences between eStation AP04 and AP05. Both use MQTT as the communication protocol, but their protocol Topics and data structure definitions are slightly different.

Difference between the AP05 Standard Edition and Developer Edition: The AP05 Developer Edition provided for system integration and the Standard Edition provided for eRetail 3.2/4.0 both use the same 4-character ID scheme, but their protocol Topics and data structure definitions differ. This document covers only the Developer Edition.

You can distinguish them by the nameplate on the back of the device.

The appearance differences among AP04 Developer Edition, AP05, and AP05 Developer Edition are shown below:

eStation Developer Edition (AP04)	ESL Access Point Model:ETAP04 Input:DC 12V  3A LOT NO:010120240307 SN/MAC:18BC5A661234  MADE IN CHINA CMIIT ID:2022DP10012 FCC ID:2ARJ5-ETAP04   211-221123   RoHS
eStation (AP05)	ESL Access Point003T Model: ETAP05 Input: DC 12V  1A CMIIT ID: 24J3213GN637 LOT NO: P20231220000133 S/N: 90A9F7400081  MADE IN CHINA    RoHS



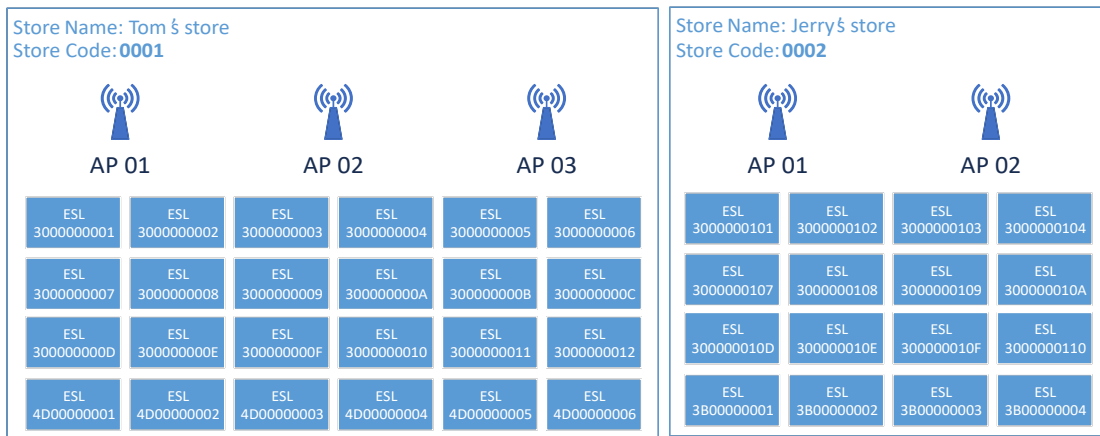
A demo project for eStation Developer Edition is available on GitHub at:
https://github.com/andersonhwang/eStation_Developer_Edition.

Note: This project is still under development. The .NET Console and WPF MVVM versions have currently been completed.

Before you start using eStation, you should understand the following key points:

1. **AP: The RF access point connected to ESLs.**
2. **AP ID: AP05 uses a 4-character ID as its globally unique identifier. This ID can be found on the nameplate attached to the back of the device.**
3. **ESL: Electronic Shelf Label; here it specifically refers to ESL Gen 3.0 using the Bluetooth Low Energy (BLE 5.0) communication protocol.**
4. **DSL: An electronic shelf label powered through a rail; currently limited to the 2.4-inch TFT ESL.**
5. **MQTT: eStation (AP05) uses the MQTT communication protocol.**
6. **Base64String: Converts an image to a Base64String. For more information about Base64String, refer to Base64 Encode and Decode - Online.**[Base64 Encode and Decode - Online](#)
7. **MessagePack: eStation (AP05) uses the MessagePack data format to reduce packet size. For more information, refer to MessagePack: It's like JSON. but fast and small. (msgpack.org).**[MessagePack: It's like JSON. but fast and small. \(msgpack.org\)](#)

For example, suppose your project has two stores, as shown below:



The data structure should be:

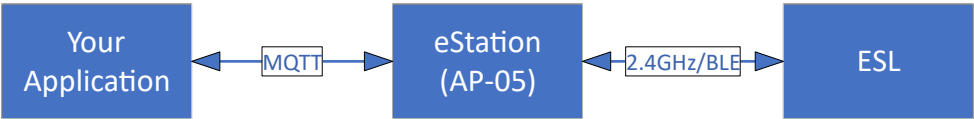
Store Code	AP ID	ESL ID
0001	01	300000000001, 300000000002, 300000000007...

0002	02	3000000000003, 3000000000004, 3000000000009 ...
	03	3000000000005, 3000000000006, 300000000000B...
	01	3000000000101, 3000000000102, 3000000000107...
	02	3000000000103, 3000000000104, 3000000000109...

Note: Normally, your application should remember the AP that successfully communicated with each ESL. Physically, however, an ESL is not bound to any AP. The ESL only logically attempts to remember the ID of the last default AP that successfully sent data to it.

1.2 eStation System Architecture

Basically, your project sends tasks containing an ESL ID and image data (Base64String) to eStation. eStation uses 2.4 GHz radio frequency (RF) to send the image data to the specified ESL and returns the result to your project. In other words, eStation sits between the application and the ESL.



This document describes two sides: your application side and the eStation device side.

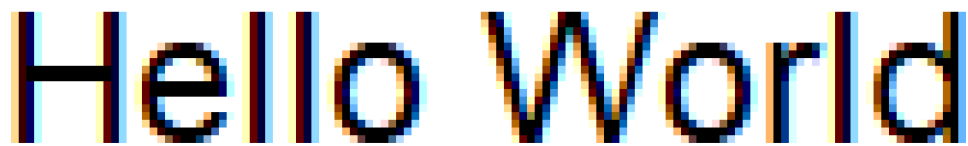
Remember that if your application and eStation are not on the same private network, you should use a strong password to protect the connection and add an X.509 certificate.

1.3 About Images

Because electronic ESL screens are pure black-and-white or black-white-red (yellow), they have no grayscale. Suppose your image looks like this:

Hello World

At 800% magnification, it looks like this:



After removing the grayscale portions, it looks like this:

Hello World

The larger the font, the more obvious the jagged edges become.

Using an ESL to display images is not recommended. If image display is required, developers should apply dithering to simulate grayscale.

One C# image-dithering algorithm is: Even more algorithms for dithering images using C# - Articles and information on C# and .NET development Topics • Cyotek[Even more algorithms for dithering images using C# - Articles and information on C# and .NET development Topics • Cyotek](#)

For example, an image may look like this:



The image results with and without dithering are shown below:



2 Using eStation

As mentioned above, there is no code-level integration with eStation. You should manage the connection and communication in your application.

There are 14 Topics: 4 are published from the eStation device side and 10 are published from the application side.

From your application side, the eStation device subscribes to the following Topics (Aliases):

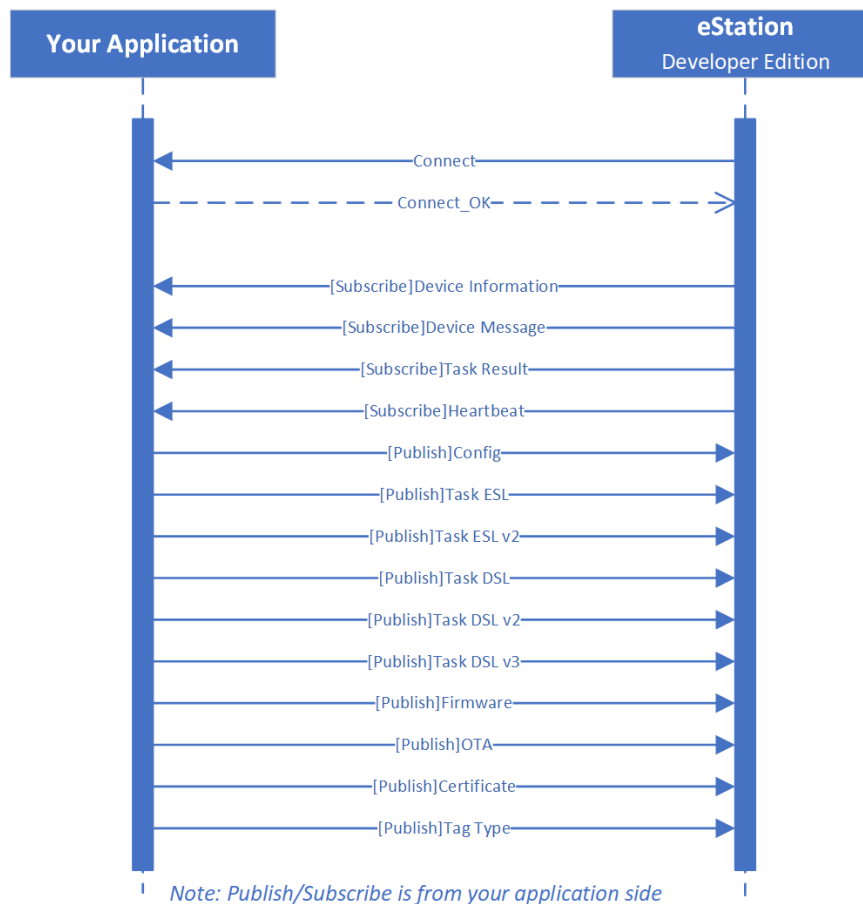
Topic	Alias	Description
configure	0x01	Use this Topic to publish eStation configuration information
taskESL	0x02	[Deprecated] Use this Topic to publish ESL tasks with image data as Base64 strings
taskESL	0x03	Use this Topic to publish ESL tasks with image data as a Byte array and an optional compression setting
taskDSL	0x04	Use this Topic to publish DSL tasks with image data as Bin data; for specific users only

taskDSL2	0x07	Use this Topic to publish DSL tasks with image data as Bin data; for specific users only
taskDSL3	0x09	Use this Topic to publish DSL tasks containing Image and Gif data; available to all users
firmware	0x05	Use this Topic to publish eStation OTA information or pre-store ESL/DSL firmware for use by the ota (0x06) Topic
ota	0x06	Use this Topic to publish ESL and DSL OTA information
cert	0x08	Use this Topic to publish certificate update information
tagType	0x0A	Use this Topic to publish ESL type updates for backward compatibility

From the eStation device side, your application should subscribe to the following Topics (Aliases):

Topic	Alias	Description
infor	0x80	Reports device information after eStation establishes a connection with the server
message	0x81	eStation returns message codes when events occur
result	0x82	eStation returns the communication result after communication with an ESL completes
heartbeat	0x83	eStation periodically returns heartbeat information

Note: You may omit the Topic Alias (default: 0). If a Topic Alias is specified, the Topic Alias takes precedence.



2.1 Connecting to eStation

eStation operates as an IoT device client and attempts to connect to your application (server side) after power-on.

The default eStation parameters are:

Default target server address	192.168.1.92:9081
Default username	test
Default password	123456

Note: Please change the username and password as soon as possible.

After connecting to an eStation, you can modify it using the configuration Topic (see 2.7 Publish Configuration Information).

However, if you cannot connect to eStation, or if you have forgotten its target server address or username/password, press and hold the reset button for a few seconds. After you see REST and the boot indication, it will restore the default target server IP address, username, and password.

Note: If you reset eStation but it still cannot connect to your application, check your firewall and Ethernet cable connection.

2.2 Receiving eStation Information

Interface Description:

Topic Alias	0x80
Topic	{Prefix}/{ID}/infor
Prefix	Topic prefix; default is "/estation" (without quotation marks)
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	Bytes of an eStationInfor object, serialized with MessagePack

eStationInfor properties:

#	Property	Type	Notes
0	ID	Char (4)	Unique 4-character ID
1	Alias	String	Alias; when it is a 2-digit number, it is shown on the digital display
2	IP	IP String	Local IP address
3	MAC	MAC String	MAC address
4	ApType	Int	Fixed at 6
5	ApVersion	String	Main application firmware version
6	ModVersion	String	Module firmware version
7	DiskSize	Int	Disk space, MB
8	FreeSpace	Int	Free space, MB
9	Server	String	Target server address; may be an IP address or domain name
10	ConnParam	String array	Connection parameters
11	Encrypt	Bool	True: enable TLS 1.2 security; False: disable it

			Disabled by default; see 2.9.2 X.509 Certificate. 2.9.2 X.509 Certificate
12	AutoIP	Bool	Whether to obtain an IP address automatically
13	LocalIP	IP String	Static IP address
14	Subnet	IP String	Subnet mask
15	Gateway	IP String	Gateway
16	Heartbeat	Int	Heartbeat interval in seconds; minimum 15

Note: When Encrypt is True, if certificates were written to the device using 0x08 before configuration, the device enters mTLS mode after configuration; the server must also support mTLS. If the device detects no usable certificate, it enters TLS mode.

2.3 Receiving eStation Messages

Interface Description:

Topic Alias	0x81
Topic	/estation/{ID}/message
ID	eStation ID
QoS Level	AtMostOnce
PayloadSegment	Bytes of an eStationMessage object, serialized with MessagePack

eStationMessage properties:

#	Property	Type	Notes
0	Code	MessageCode	Enum type

MessageCode enumeration values:

0. OK General response
1. Idle
2. Result Communication result returned
3. Heartbeat Heartbeat monitoring feedback
4. ModError Communication module error
5. AppError Main application error
6. Busy Device busy
7. MaxLimit The current data queue has reached its limit (more than 10,000 pending items or memory usage exceeds 512 MB)
8. InvalidTaskESL Invalid ESL task data
9. InvalidTaskDSL Invalid DSL task data
10. InvalidConfig Invalid configuration data
11. InvalidOTA Invalid OTA data

2.4 Receiving eStation Results

Interface Description:

Topic Alias	0x82
Topic	/estation/{ID}/result

ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	Bytes of a TaskResult object, serialized with MessagePack TaskResult contains a list of TagResult objects

TaskResult properties:

#	Property	Type	Notes
0	Port	Int	MOD serial port number
1	WaitCount	Int	Total number of tasks in the cache
2	SendCount	Int	Number of tasks being sent (in the RF module)
3	Message	String	Reserved field
4	Tags	List<TagResult>	ESL result list

TagResult properties:

#	Property	Type	Notes
0	TagID	String	ESL ID
1	RfPower	Int	RF power, dBm; -256 means no data
2	Battery	Byte	Battery voltage, V; 0 means no data
3	Version	Byte	ESL firmware version; reserved
4	Status	Byte	Status; reserved
5	Token	Int	Token, used for data-loop closure; see 2.5 Token
6	Temperature	Int	Temperature, °C; 0 means no data
7	Channel	Int	Channel; reserved
8	UtcTime	Byte array	UTC time; reserved; DSL only
9	TimePercent	Byte array	Time confidence; reserved; DSL only
10	Count	Byte	Counter; reserved; DSL only

Note: There is a queue inside the AP. When the application sends task data to the AP, it is sent to the RF module immediately if the RF module is idle; otherwise it waits until the RF module becomes idle. Therefore, communication between the application and eStation is asynchronous.

2.5 Receiving eStation Heartbeats

Interface Description:

Topic Alias	0x83
Topic	/estation/{ID}/heartbeat
ID	eStation ID
QoS Level	AtMostOnce
PayloadSegment	Bytes of an ApHeartbeat object, serialized with MessagePack ApHeartbeat contains a list of TagHeartbeat objects

ApHeartbeat properties:

#	Property	Type	Notes
0	Id	String	AP ID
1	ConfigVersion	Int	Configuration version (default 0; reserved)
2	ApVersion	String	AP version

3	ModVersion	String	Bluetooth module version
4	Message	MessageCode	Message; see 2.3 MessageCode
5	MessageEx	String	Message (extended field)
6	WaitCount	Int	Number of ESLs currently queued
7	SendCount	Int	Number of ESLs currently communicating
8	Tags	TagHeartbeat list	ESL heartbeat information

TagHeartbeat properties:

#	Property	Type	Notes
0	TagId	String	ESL/DSL ID
1	RfPower	Int	RF power, dBm; -256 means no data
2	Battery	Byte	Battery voltage, V; 0 means no data
3	Version	Byte	ESL firmware version; reserved
4	Status	Byte	Status; reserved
5	Token	Int	Token, used for data-loop closure; see 2.5 Token
6	Temperature	Int	Temperature, °C; 0 means no data
7	Channel	Int	Channel; reserved
8	UtcTime	Byte array	UTC time; DSL only
9	TimePercent	Byte	Time confidence; DSL only
10	Count	Byte	Counter; DSL only
11	Factory	Byte	Factory code, used for OTA
12	Color	Byte	Screen color code, used for OTA
13	Size	Byte	Screen size code, used for OTA
14	Type	Byte	Screen type code, used for OTA

Note: The heartbeat and result data structures are similar. A heartbeat contains ESL heartbeat information detected by the AP, whereas a result is returned after data is issued by your application. The former occurs proactively; the latter occurs in response to a task.

2.6 Publishing Configuration Information

Interface Description:

Topic Alias	0x01
Topic	/estation/{ID}/ config
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	Bytes of an eStationConfig object, serialized with MessagePack

eStationConfig properties:

#	Property	Type	Notes
0	Alias	String	Alias; when it is a 2-digit number, it is shown on the digital display
1	Server	String	Target server address
2	ConnParam	String array	Connection parameters
3	Encrypt	Bool	True: enable TLS 1.2 security; False: disable

			it Disabled by default; see 2.9.2 X.509 Certificate 2.9.2 X.509 Certificate
4	AutoIP	Bool	Whether to obtain an IP address automatically
5	LocalIP	IP String	Static IP address
6	Subnet	IP String	Subnet mask
7	Gateway	IP String	Gateway
8	Heartbeat	Int	Heartbeat interval in seconds; minimum 15

Note: This Topic also supports `/estation/{ID}/configure`.

2.7 Publishing ESL Tasks (Base64 Version)

Note:

1. This is a deprecated Topic. The Bytes-version Topic in Section 2.8 is recommended.
2. Only 3-color (Black-White-Red) and 4-color (Black-White-Red-Yellow) ESLs are supported.

Interface Description:

Topic Alias	0x02
Topic	/estation/{ID}/taskESL
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	A list array of ESLEntity objects, serialized with MessagePack

ESLEntity properties:

#	Property	Type	Notes
0	TagID	String	ESL ID
1	Pattern	Int, Pattern	Pattern code; see 3.1 Pattern 3.1 Pattern
2	PageIndex	Int, PageIndex	Page index; see 3.2 PageIndex 3.2 PageIndex
3	R	Bool	Red LED
4	G	Bool	Green LED
5	B	Bool	Blue LED
6	Times	Int, 2-byte range	Number of LED flashes, seconds
7	Token	Int, 2-byte range	Token, returned in 2.3 Token 2.3 Token
8	CurrentKey	Int, 8-byte range	Current key; leave empty when using the default key
9	NewKey	8 bytes	New key; leave empty if no change is required
10	Base64String	String	Base64String value of the image in Bitmap format

Overwrite: For an ESL ID, if your application sends another task without waiting for the previous task result, eStation discards the previous task.

Pixels: The pixel format must be BGRA; otherwise color parsing errors or display abnormalities may occur.

Size: eStation checks the size (length) of the image data (Base64String) in an ESL task. If the length is incorrect (or the actual image dimensions do not match the ESL screen dimensions/padded dimensions), the image cannot be updated successfully. The actual image dimensions cannot be inferred from the image Byte array for padding; for example, 300×400 and 200×600 Bitmap images have the same data length.

For the relationship between ESL type and screen resolution, see 3.1 ESL Gen 3.0 Model List.

In particular, because the screen data width is a multiple of 8 while some models may not meet this requirement, data may be sent using the actual screen dimensions or padded to a multiple of 8 (the extra portion is neither processed nor displayed). For example, for the ET0213 ESL, both 250×122 and 256×122 images are accepted, but only the 250×122 portion is displayed.

Limit: When data is pushed as Base64 strings, the eStation task-cache limit is relatively low (about 800 tasks in a test with 4.2-inch 3-color ESLs). To increase the limit, use the Bytes version for pushing data.

Key: For more information about CurrentKey and NewKey, see 2.12 Secure Communication.[2.12 Secure Communication](#)

2.8 Publishing ESL Tasks (Bytes Version)

Interface Description:

Topic Alias	0x03
Topic	/estation/{ID}/taskESL2
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	A list array of ESLEntity2 objects, serialized with MessagePack

ESLEntity2 properties:

#	Property	Type	Notes
0	TagID	String	ESL ID
1	Pattern	Int, Pattern	Pattern code; see 3.1 Pattern 3.1 Pattern
2	PageIndex	Int, PageIndex	Page index; see 3.2 PageIndex 3.2 PageIndex
3	R	Bool	Red LED
4	G	Bool	Green LED
5	B	Bool	Blue LED
6	Times	2 bytes	Number of LED flashes, seconds
7	Token	2 bytes	Token, returned in 2.3 Token 2.3 Token
8	CurrentKey	8 bytes	Current key; leave empty when using the default key
9	NewKey	8 bytes	New key; leave empty if no change is required
10	Bytes	Byte array	Image data (see description below)

11	Compress	Bool	Whether to compress; default True; GZip compression
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Pixels: This Topic has the same image-pixel requirements as 2.7 Publishing ESL Tasks (Base64 Version).

Size: This Topic has the same image-size requirements as 2.7 Publishing ESL Tasks (Base64 Version).

Limit: With the Bytes-version Topic, the eStation limit is currently 20,000 tasks (sufficient for most scenarios). Tasks exceeding this limit are discarded, and MessageCode.MaxLimit is returned through the Topic described in 2.3 Receiving AP Messages.

Image data: Bitmap pixel arrays (BGRA32 and BGR24) are supported, as are image files such as Bmp, Jpeg, and Png. GZip compression is supported; in this case Compress must be set to True. Unlike taskESL, image dimensions must strictly match the screen pixel dimensions defined for the product model.

2.9 Publishing DSL Tasks

Interface Description:

Topic Alias	0x04
Topic	/estation/{ID}/taskDSL
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	A list array of DSLEntity objects, serialized with MessagePack

DSLEntity properties:

#	Property	Type	Notes
0	TagID	String	ESL ID
1	R	Bool	Red LED
2	G	Bool	Green LED
3	B	Bool	Blue LED
4	Period	2 bytes	Number of LED flashes, -1–3600; default 3600; unit: times
5	Interval	2 bytes	LED interval, 100–10000; default 1000; unit: ms
6	Duration	8 bytes	LED on-time, 50–100; default 50; unit: ms
7	Token	8 bytes	Token, returned in 2.3 Token 2.3 Token
8	HexData	Byte array	Bin file data stream (for specific customers)

Data: A Bin file is an image/instruction file specifically generated for the 2.4-inch TFT ESL (DSL).

Steady on: When Period = -1, the LED remains steadily on (Interval and Duration are ignored).

2.10 Publishing DSL Tasks (with Pattern and Key)

Interface Description:

Topic Alias	0x07
Topic	/estation/{ID}/taskDSL2
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	A list array of DSLEntity2 objects, serialized with MessagePack

DSLEntity2 properties:

#	Property	Type	Notes
0	TagID	String	ESL ID
1	R	Bool	Red LED
2	G	Bool	Green LED
3	B	Bool	Blue LED
4	Period	2 bytes	Number of LED flashes, -1–3600; default 3600; unit: times
5	Interval	2 bytes	LED interval, 100–10000; default 1000; unit: s
6	Duration	8 bytes	LED on-time, 50–100; default 50; unit: ms
7	Token	8 bytes	Token, returned in 2.3 Token 2.3 Token
8	HexData	Byte array	Bin file data stream
9	Pattern	Int, Pattern	Pattern code; see 3.1 Pattern 3.1 Pattern
10	CurrentKey	8 bytes	Current key; leave empty when using the default key
11	NewKey	8 bytes	New key; leave empty if no change is required

Data: A Bin file is an image/instruction file specifically generated for the 2.4-inch TFT ESL (DSL).

Steady on: When Period = -1, the LED remains steadily on (Interval and Duration are ignored).

2.11 Publishing DSL Tasks (for Image and Gif)

Interface Description:

Topic Alias	0x09
Topic	/estation/{ID}/taskDSL3
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	A list array of DSLEntity3 objects, serialized with MessagePack

DSLEntity3 properties:

#	Property	Type	Notes
0	TagID	String	ESL ID
1	R	Bool	Red LED
2	G	Bool	Green LED

3	B	Bool	Blue LED
4	Period	2 bytes	Number of LED flashes, -1–3600; default 3600; unit: times
5	Interval	2 bytes	LED interval, 100–10000; default 1000; unit: s
6	Duration	8 bytes	LED on-time, 50–100; default 50; unit: ms
7	Token	8 bytes	Token, returned in 2.3 Token 2.3 Token
8	HexData	Byte array	Image/Gif file data stream
9	Pattern	Int, Pattern	Pattern code; see 3.1 Pattern 3.1 Pattern
10	CurrentKey	8 bytes	Current key; leave empty when using the default key
11	NewKey	8 bytes	New key; leave empty if no change is required
12	IsGif	Bool	Whether HexData contains Gif data
13	Width	Int	Image/Gif Width
14	Height	Int	Image/Gif Height
15	Top	Int	IsGif = true, Gif top offset
16	Left	Int	IsGif = true, Gif left offset

Note: When pushing an Image, Width is fixed at 320 and Height at 240. When pushing a Gif, neither Width nor Height may exceed 100 pixels, and the number of Frames may not exceed 8.

2.12 Publishing OTA Tasks (Firmware)

Interface Description:

Topic Alias	0x05
Topic	/estation/{ID}/firmware
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	A list array of OTAData objects, serialized with MessagePack

OTAData properties:

#	Property	Type	Notes
0	DownloadUrl	String	Firmware download URL
1	ConfirmUrl	String	Target server address
2	Type	Int	OTA type: 0-eStation, 1/2-MOD, 3-ESL/DSL
3	Version	String	Version number
4	Name	String	Firmware name
5	MD5	IP String	Static IP address

Note:

- For this Topic, Type=0 is treated as eStation firmware; after download verification, eStation immediately restarts and performs the OTA. Type=1 or 2 is treated as MOD firmware and immediately starts a MOD OTA. Type=3 is treated as pre-stored ESL/DSL firmware for the ESL OTA described in Section 2.11.
- Once OTA starts, do not communicate with eStation until the OTA has definitely completed.

- MD5 format: In earlier versions (<1.0.28), the format is 00-11-22-33-44-55-66-77. In later versions (>=1.0.28), the format is 00111223344556677.

2.13 Publishing OTA Tasks (ESL)

Interface Description:

Topic Alias	0x06
Topic	/estation/{ID}/ota
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	A list array of OTATask objects, serialized with MessagePack

OTATask properties:

#	Property	Type	Notes
0	Step	Int	Type: 0-PreOTA; 1-OTA; 2-Clean
1	Firmware	String	Firmware file name
2	TagIDList	String array	List of ESL IDs to update

Note:

- This Topic depends on the firmware described in Section 2.10; otherwise the firmware cannot be found for updating.
- This Topic must be executed at least three consecutive times (Step = 0/1/2 in sequence). Based on the results in Section 2.4, verify that all ESLs in the current Step have communicated successfully before proceeding to the next Step.
- The TagIDList in this Topic must contain ESL IDs of the same model, and Firmware must be the 2-character code for that model.

2.14 Publishing Certificate Import Tasks

Interface Description:

Topic Alias	0x08
Topic	/estation/{ID}/cert
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	An ApCert object, serialized with MessagePack

ApCert properties:

#	Property	Type	Notes
0	DownloadUrl	String	Download URL
1	ConfirmUrl	String	Confirm URL
2	CertName	String	Client certificate name
3	KeyName	String	Client key name
4	CustomTrustStore	String array	Custom trust store certificate(s)
5	ExtraStore	String array	Extra store certificate(s)
6	MD5	String	MD5 of the certificate ZIP archive

Note:

1. This Topic only imports certificates to the device; it does not enable mTLS communication. To enable mTLS, after pushing the certificates with this Topic, set Encrypt=true using the Config Topic (0x00).
2. Generating the Client certificate and key on the server and then downloading them to the device is not best practice. For security, DownloadUrl access should be one-time and its validity period should be short.
3. Certificate package: Package the certificate files into a ZIP archive.

2.15 Publishing ESL Type Information

Interface Description:

Topic Alias	0x0A
Topic	/estation/{ID}/tagType
ID	eStation ID
QoS Level	ExactlyOnce
PayloadSegment	An array of TagType objects, serialized with MessagePack

TagType properties:

#	Property	Type	Notes
0	Code	String	ESL type code (first 2 characters of the ESL ID)
1	Height	Int	Height
2	Width	Int	Width
3	Type	String	ESL type
4	Color	Enum	Color type; see 3.5 ColorType for definitions 3.5 ColorType

Note:

1. Do not modify ESL type parameters arbitrarily. Incorrect ESL type parameters can cause display errors.
2. Use the correct data source when updating ESL types. The sole data source is: https://github.com/andersonhwang/eStation_Developer_Demo/blob/main/TagType.e.json
3. For an existing Code, this interface overwrites the previous data.

2.16 Secure Communication

2.12.1. ESL Key

Generally, if another project uses its eStation (AP) to control your project's ESLs, unexpected and unacceptable results may occur. In this case, it is recommended that you set a key for your ESLs.

When you receive new ESLs from the sales manager, the default ESL key is empty (11 bytes, i.e. FFFFFFFFFFFFFFFF). You may keep it or set CurrentKey and NewKey. Subsequent communication should use the new key.

1. The key has a fixed length of 22 characters (11 bytes).
2. Only hexadecimal characters (0123456789ABCDEF) are accepted.
3. If CurrentKey is empty, eStation substitutes the default key.
4. If New Key is empty, the AP does not change the key. If New Key is not empty, the AP changes the key.
5. **Very important: Store your key securely. If you forget the key, you will lose control of the ESL.**
6. Only ESLs with firmware versions later than 35 support key-based communication.

2.12.2. X.509 Certificate

If the TLS 1.2 security protocol is enabled in 2.7 Publishing Configuration Information, the server can use a custom X.509 certificate to encrypt communication data. How the server uses the X.509 certificate and TLS 1.2 depends on the server-side language/framework selected by the developer.

If the server is developed using C# and MQTTnet, a PFX certificate is required. Example code follows:

```
public async Task Run()
{
    var options = new MqttServerOptionsBuilder()
        .WithDefaultEndpoint()
        .WithEncryptedEndpoint()
        .WithEncryptedEndpointPort(PORT) // TCP port in your server side
        .WithEncryptionCertificate(CERTIFICATE_PATH) // Your certificate path
        .WithEncryptionSslProtocol(SslProtocols.Tls12)
        .Build();
    _server = _factory.CreateMqttServer(options);
    _server.ValidatingConnectionAsync += ServerOnValidatingConnectionAsync;
    _server.ClientConnectedAsync += ServeOnClientConnectedAsync;
    _server.ClientDisconnectedAsync += ServerOnClientDisconnectedAsync;
    _server.InterceptingPublishAsync += ServerOnInterceptingPublishAsync;
    await _server.StartAsync();
}
```

3 Reference

3.1 ESL Gen 3.0 Model List

Model	Size (inches)	Screen	Color	Pixels (H*W)	Freezer
ET0154-33	1.54	Eink	B/W/R	200*200	-
ET0213-36	2.13	Eink	B/W/R	250*122	-
ET0213-39	2.13	Eink	B/W	250*122	Y
ET0266-3A	2.66	Eink	B/W/R	296*152	-

ET0266-5B	2.66	Eink	B/W	296*152	Y
ET0290-3D	2.9	Eink	B/W/R	296*128	-
ET0290-3F	2.9	Eink	B/W	296*128	-
ET0290-54	2.9	Eink	B/W	296*128	Y
ET0420-40	4.2	Eink	B/W/R	400*300	-
ET0420-42	4.2	Eink	B/W	400*300	-
ET0750-44	7.5	Eink	B/W/R	800*480	-
ET0750-46	7.5	Eink	B/W	800*480	-
ET0430-4C	4.3	Eink	B/W/R	522*152	-
ET0580-4F	5.8	Eink	B/W/R	648*480	-
ET0350-55	3.5	Eink	B/W/R	384*184	-
ET1250-58	12.5	Eink	B/W/R	1304*984	-
ET0420-5D	4.2	TFT	B/W/R	400*300	-
ET1020-64	10.2	Eink	B/W/R	960*640	-
ET1020-67	10.20	Eink	B/W	960*640	-
ET1330-68	13.3	Eink	B/W/R	960*680	-
ET0579-6F	5.8	Eink	B/W/R	792*272	-
ET0154H-71	1.54	Eink	B/W/R	200*200	-
ET0130H-73	1.3	Eink	B/W/R	200*144	-
ET0154-80	1.54	Eink	B/W/R/Y	200*200	-
ET0213-81	2.13	Eink	B/W/R/Y	250*122	-
ET0266-82	2.66	Eink	B/W/R/Y	296*152	-
ET0266-83	2.66	Eink	B/W/R/Y	360*184	-
ET0290-84	2.9	Eink	B/W/R/Y	296*128	-
ET0290-85	2.9	Eink	B/W/R/Y	384*168	-
ET0350-86	3.5	Eink	B/W/R/Y	384*184	-
ET0420-87	4.2	Eink	B/W/R/Y	400*300	-
ET0580-88	5.8	Eink	B/W/R/Y	648*480	-
ET0750-89	7.5	Eink	B/W/R/Y	800*480	-
ET0750-8A	7.5	Eink	B/W/R/Y	880*528	-
ET1020-8B	10.2	Eink	B/W/R/Y	960*640	-
ET0097H-8C	0.97	Eink	B/W/R/Y	184*88	-
ET0130H-8D	1.3	Eink	B/W/R/Y	200*144	-
ET1330-8E	13.3	Eink	B/W/R/Y	960*680	-
ET0370-8F	3.7	Eink	B/W/R/Y	416*240	-
ET0154H-90	1.54	Eink	B/W/R/Y	200*200	-
ET0430-91	4.3	Eink	B/W/R/Y	522*152	-
ET0213-95	2.13	Eink	4-Color	250*122	Y

ET0266-96	2.66	Eink	4-Color	296*152	Y
ET0290-97	2.9	Eink	4-Color	296*128	Y
ET0600-99	5.8	Eink	B/W/R/W	792*272	-
ET0730-A0	7.3	Eink	6-Color	800*480	-
ET1330-A1	13.3	Eink	6-Color	1600*1200	-
ET0240C-D0	2.4	TFT	Full Color?	320*240	-

Note:

1. In the Color column, B means Black, W means White, R means Red, and Y means Yellow.
2. In the Color column, 6-Color means Black, White, Red, Yellow, Green, and Blue.

3.2 Pattern

Pattern	Description
0- UpdateDisplay	Update and display
1- Update	Update without display
2- Display	Display
3- LED	Flash LED
4- Key	Update key (see 2.12.1 ESL Key) 2.12.1 ESL Key
5- ClearGif	Clear Gif (DSL only)
6- ClearGifUpdateImage	Clear Gif and update static image (DSL only)

3.3 PageIndex

Page ID	Description
0- P0	Page 1
1- P1	Page 2
2- P2	Page 3
3- P3	Page 4
4- P4	Page 5; compressed image size must not exceed 32 KB
5- P5	Page 6; compressed image size must not exceed 32 KB
6- P6	Page 7; compressed image size must not exceed 32 KB
7- P7	Page 8; compressed image size must not exceed 32 KB

Note: Currently, all ESLs have an 8-page data cache. Due to storage limitations, images sent to pages 5-8 must not exceed 32 KB after compression (LZSS compression is implemented inside the AP; the server still sends the original image).

3.4 RGB Color Definition

The ESL LED uses an RGB tri-color LED and can visually produce seven colors.

Color	R	G	B
Off	×	×	×
Blue	×	×	√
Green	×	√	×
Cyan	×	√	√
Red	√	×	×
Purple	√	×	√
Yellow	√	√	×
White	√	√	√

3.5 ColorType

ColorType	Description
1 – Black	Black
2 – Red	Red
4 – Yellow	Yellow
3 – Black Red	Black, Red
5 – Black Yellow	Black, Yellow
7 – Black Red Yellow	Black, Red, Yellow
8 – Color4	Black, Red, Yellow
9 – Color6	Black, Yellow, Red, Orange(X), Blue, Green
10 – Color7	Black, Yellow, Red, Dark Orange(X), Dark Gray, Light Gray, Bright Orange(X)

Note:

1. White is the default color.
2. The orange colors in Color6 and Color7 are currently unused.

